

PESKINE AND SZPIRO'S DIMENSION INEQUALITY HOLDS OVER LOCAL RINGS OF DEPTH AT MOST ONE

LARS WINTHER CHRISTENSEN AND SAEED NASSEH

ABSTRACT. We prove that the general version—the one that includes the dimension of the tensor product—of Peskine and Szpiro's Dimension Inequality holds over noetherian local rings of depth at most one.

Let R be a commutative noetherian local ring. Serre [7] proved that if R is regular and M and N are finitely generated R -modules, then the inequality

$$\dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R N)$$

holds. It was conjectured by Peskine and Szpiro [6] that this inequality also would hold over a singular local ring, as long as one of the modules, say M , has finite projective dimension. In this paper, we will refer to this inequality as the *Dimension Inequality*, though some authors reserve that name for the special case where the R -module $M \otimes_R N$ has finite length and hence dimension 0.

An example by Ma [4, §2.3] shows that the Dimension Inequality does not hold in general. To be precise, Ma constructs a Cohen–Macaulay local ring of dimension 3 over which the inequality fails. In that light, one asks under what extra assumptions on the ring the inequality is valid. Our contribution is the following.

Theorem 1. *Let R be a commutative noetherian local ring and M and N be finitely generated R -modules. If $\mathrm{pd}_R M$ is finite and $\mathrm{depth} R \leq 1$ holds, then one has*

$$(1) \quad \dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R N).$$

Proof. By Peskine and Szpiro's Intersection Theorem, the inequality

$$(2) \quad \dim_R N \leq \mathrm{pd}_R M + \dim_R(M \otimes_R N)$$

holds. It follows from the Auslander–Buchsbaum Formula and the assumptions on M and R that $\mathrm{pd}_R M$ is 0 or 1. In the former case, M is free, so $\dim_R M = \dim R$ holds and thus (1) follows from (2) by adding this quantity to both side.

Now assume that $\mathrm{pd}_R M = 1$ holds and consider the minimal free resolution

$$0 \longrightarrow R^a \longrightarrow R^b \longrightarrow M \longrightarrow 0$$

where $a \leq b$ are positive integers. For every minimal prime ideal $\mathfrak{p}$ in R it induces an exact sequence

$$(3) \quad 0 \longrightarrow R_{\mathfrak{p}}^a \longrightarrow R_{\mathfrak{p}}^b \longrightarrow M_{\mathfrak{p}} \rightarrow 0$$

of finite length $R_{\mathfrak{p}}$ -modules.

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If $a = b$ holds, then (3) yields $\text{length}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 0$ and thus $M_{\mathfrak{p}} = 0$ for every minimal prime ideal $\mathfrak{p}$ of R . It follows that $\dim_R M \leq \dim R - 1$ holds, and adding this to (2) yields:

$$\begin{aligned} \dim_R M + \dim_R N &\leq \dim R - 1 + \text{pd}_R M + \dim_R(M \otimes_R N) \\ &= \dim R + \dim_R(M \otimes_R N). \end{aligned}$$

If $a < b$ holds, then (3) yields

$$\text{length}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = (b - a) \text{length}_{R_{\mathfrak{p}}}(R_{\mathfrak{p}}) > 0$$

and, therefore, $M_{\mathfrak{p}} \neq 0$ for every minimal prime ideal $\mathfrak{p}$ in R . Thus, $\text{Supp}_R M$ is all of $\text{Spec } R$, whence one has

$$\text{Supp}_R(M \otimes_R N) = \text{Supp}_R M \cap \text{Supp}_R N = \text{Supp}_R N.$$

In particular, the equalities $\dim_R M = \dim R$ and $\dim_R N = \dim_R(M \otimes_R N)$ hold, whence equality holds in (1). $\square$

Remark 2. The assumption on the projective dimension of M in Theorem 1 can not be dropped; see [2, Exa. 18.5.4]. Also, the proof shows that, the assumption $\text{depth } R \leq 1$ can be replaced by the weaker assumption $\text{pd}_R M \leq 1$.

Theorem 1 extends to complexes as follows:

Corollary 3. *Let R be a commutative noetherian local ring and M and N be R -complexes with finitely generated homology modules and $H_i(M) = 0 = H_i(N)$ for $i \ll 0$. If $\text{pd}_R(H_i(M))$ is finite for all $i \in \mathbb{Z}$ and $\text{depth } R \leq 1$ holds, then one has*

$$\dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R^{\mathbf{L}} N).$$

Proof. The proof of [2, Thm. 20.1.34] applies verbatim: It is a standard reduction to the module case and the assumption on M ensures that Theorem 1 applies. $\square$

Remark 4. If the R -complex M in Corollary 3 has bounded homology, then the assumption on the homology modules implies that M has finite projective dimension; see for example Aihara et. al [1, Thm. 4.1]

This project arose from the question, does the Dimension Inequality hold over fiber product rings? As the next example shows, the answer is affirmative for a particular class of fiber products. These fiber product rings are precisely the local rings with decomposable maximal ideal; see Ogoma [5, Lem. 3.1].

Example 5. Let $(S, \mathfrak{s}, k)$ and $(T, \mathfrak{t}, k)$ be commutative noetherian local rings such that $S \neq k \neq T$ and let $R = S \times_k T$ be the fiber product of S and T over k , i.e.

$$R = \{(s, t) \in S \times T \mid \pi_S(s) = \pi_T(t)\}$$

where $S \xrightarrow{\pi_S} k \xleftarrow{\pi_T} T$ are the natural surjections. It is a commutative noetherian local ring with $\mathfrak{m} = \mathfrak{s} \oplus \mathfrak{t}$ and residue field k . By work of Lescot [3] one has

$$\text{depth } R = \min\{\text{depth } S, \text{depth } T, 1\}.$$

In particular, $\text{depth } R \leq 1$ holds and, therefore, Theorem 1 applies.

Remark 6. The ring considered in [2, Exa. 18.5.14]—already mentioned in Remark 2—is the fiber product $k[[x]] \times_k k[[y]]$. Thus, the assumption on the projective dimension of M can not be dropped for fiber products.

The local ring in Ma's [4, §2.3] counterexample to the Dimension Inequality is also a fiber product but, of course, different from those considered above: Ma's ring $A \times_C B$ is glued over a ring C of dimension 2 rather than over the residue field.

Here is a local ring that is covered by Theorem 1 and not a fiber product:

Example 7. The ring $R = k[[x, y]]/(y^2 - x^3)$ is a Cohen–Macaulay local domain (hence not a fiber product) with $\dim R = 1$, so Theorem 1 applies to R . Set $M = R/(x)$ and $N = \mathfrak{m}$. As x is not a zerodivisor, one has $\mathrm{pd}_R M = 1$ and $\dim_R M = 0 = \dim_R(M \otimes_R N)$. Since R is a domain, $\dim_R N = \dim R = 1$ holds. Thus equality holds in the Dimension Inequality (1) as well as in the inequality (2).

In closing we note that the proof of [4, Prop. B.5(1)] adapts to show that also the general version of the Dimension Inequality holds for rings of dimension at most 2.

Proposition 8. *Let R be a commutative noetherian local ring and M and N be finitely generated R -modules. If $\mathrm{pd}_R M$ is finite and $\dim R \leq 2$ holds, then one has*

$$\dim_R M + \dim_R N \leq \dim R + \dim_R(M \otimes_R N) .$$

Proof. First note that by Theorem 1 we can assume that $\dim R = 2$ holds.

If $\dim_R M = 0$ holds, then the asserted inequality holds trivially. If $\dim_R M = 1$ holds, then there is, as in the proof of [4, Prop. B.5], a non-zero-divisor $z \in R$ with $V(z) \subseteq \mathrm{Supp}_R M$ and, therefore,

$$\mathrm{Supp}_R(N/zN) = \mathrm{Supp}_R N \cap V(z) \subseteq \mathrm{Supp}_R N \cap \mathrm{Supp}_R M = \mathrm{Supp}_R(M \otimes_R N) .$$

Thus, $\dim_R(N/zN) \leq \dim_R(M \otimes_R N)$ holds, whence one has

$$\dim_R N \leq 1 + \dim_R(N/zN) \leq 1 + \dim_R(M \otimes_R N) .$$

If $\dim_R M = 2$ holds, then $\mathrm{Supp}_R M = \mathrm{Spec} R$, see again the proof of [4, Prop. B.5], and thus $\mathrm{Supp}_R(M \otimes_R N) = \mathrm{Supp}_R N$ holds. In particular, one has $\dim_R N = \dim_R(M \otimes_R N)$, and the asserted inequality follows; actually equality holds. $\square$

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TEXAS TECH UNIVERSITY, LUBBOCK, TX 79409, U.S.A.

Email address: `lars.w.christensen@ttu.edu`

URL: `https://larswinther.github.io/homepage/`

DEPARTMENT OF MATHEMATICAL SCIENCES, GEORGIA SOUTHERN UNIVERSITY, STATESBORO,
GA 30460, U.S.A.

Email address: `snasseh@georgiasouthern.edu`

URL: `https://cosm.georgiasouthern.edu/math/saeed.nasseh`